

PLACEMENT SKILL-GAP HUB

Technical Implementation Plan & Walkthrough

Hackathon
HACKORBIT 2K26

Team
Codeavengers

Track
Track A — DataDrishti

Institution: Bannari Amman Institute of Technology

Document status. This is a build specification, not a completed-results report — the dataset and dashboard described here are the plan the team will implement. All figures shown as examples in this document are **Illustrative / Sample Data** and are explicitly labeled as such. None of them are real Bannari Amman Institute of Technology placement statistics. For the hour-by-hour build order, see the companion document **workflow_HACKORBIT.pdf**.

1. Executive Summary

The Placement Skill-Gap Hub is a Power BI solution built for HACKORBIT 2K26's DataDrishti track. Rather than reporting placement outcomes alone, it connects three data layers — placement results, industry skill demand, and student skill readiness — into a single measure: the **skill gap**. That gap is classified by priority and converted into a plain-language training recommendation for each skill. The result is a dashboard that does not just describe what happened in past placements; it points at what an institution should do next.

The build is deliberately scoped to what two people can finish, test, and defend in two days: one three-table dataset, three dashboard pages, ten core DAX measures, and two signature visuals (the Skill Gap Priority Matrix and a rule-based Training Recommendation table). Everything in this document is written to survive a skeptical judge asking exactly where each number came from.

2. Problem Statement & Winning Thesis

Official problem statement (Track A — DataDrishti):

“Placement Skill-Gap Hub — Model placement stats across branches.”

Our central thesis:

“Placement statistics tell us WHAT happened. Skill-gap intelligence tells us WHY — and WHAT TO IMPROVE.”

A standard placement dashboard stops at descriptive metrics — placement rate, average package, companies visited. Those numbers say nothing about why a student wasn't placed or what a training team should do differently next semester. Our dashboard extends the chain one step further, into cause and action:



The question this dashboard is built to answer, and the question every visual on it should trace back to: **“Which skills should this institution prioritize to improve placement readiness?”**

3. Solution Architecture & Differentiation

A conventional placement dashboard shows placement percentage, average package, company visits, and branch comparisons. That is necessary but purely descriptive — it is not, by itself, a differentiator in a room full of similar dashboards. Our differentiation is not a new chart type; it is a different question being answered.

Ordinary placement dashboard	Placement Skill-Gap Hub
“How many students got placed?”	“Why weren’t the rest placement-ready, and what should change?”
Reports outcomes after the fact	Connects outcomes to the skill gaps that likely caused them
One data layer (placement records)	Three linked layers: outcomes, industry demand, student readiness
Ends at a chart	Ends at a rule-based, per-skill training recommendation

Defensible positioning statement. We do not claim this has never been built before — that is unprovable and an easy attack surface. Instead: “Our differentiation is the integration of placement outcomes, recruiter skill demand, and student readiness into a single actionable skill-gap workflow.” This is specific, true, and demonstrable live.

4. Dataset Architecture

Sample / Demonstration Dataset. No real Bannari Amman Institute of Technology placement, student, or recruiter data is used. All records below are synthetically generated with deliberate, logical relationships (e.g. readiness varies believably by branch; frequently-required skills appear more often in the demand table) so the dashboard's behaviour is realistic without claiming to be real.

Three tables, not five — enough to form a clean, explainable star-schema-like model, and small enough to build and debug within the two-day window.

Table 1 — Students_Placement (one row per student)

Field	Type	Notes
Student_ID	Text (key)	Unique identifier, e.g. STU0001
Branch	Text	CSE, AIDS, ECE, MECH, etc. — standardized spellings
Batch	Text/Number	Graduating year, e.g. 2026
Placement_Status	Text	“Placed” or “Not Placed”
Company	Text	Blank if not placed
Package	Number (LPA)	Blank if not placed

Table 2 — Student_Skills (one row per student-skill pair)

Field	Type	Notes
Student_ID	Text (foreign key)	Links to Students_Placement
Skill	Text	e.g. Python, SQL, Power BI, Machine Learning — matches Table 3
Assessment_Score	Number (0–100)	Proxy for individual skill readiness

Table 3 — Industry_Skill_Demand (one row per skill)

Field	Type	Notes
Skill	Text (key)	Matches Student_Skills[Skill] exactly
Demand_Level	Number (0–100)	Recruiter-requirement intensity for this skill, as a percentage
Companies_Requiring_Count	Number	How many sample companies listed this skill as required

Why not five tables (DIM_STUDENT / FACT_PLACEMENT / DIM_SKILL / FACT_STUDENT_SKILL / FACT_SKILL_DEMAND)? That structure is more “textbook,” but for a 2-day build it adds join complexity without adding analytical power we’d actually use. Three tables reach the same measures with fewer relationships to debug and an easier model to explain on stage.

5. Data Model

The model is a simple one-directional chain, not a full star schema — appropriate for three tables and a scatter/matrix-driven dashboard rather than a deep enterprise model.

Relationship	Cardinality	Direction	Purpose
Students_Placement[Student_ID] → Student_Skills[Student_ID]	1 : Many	Single (filters flow Placement → Skills)	Lets Branch/Batch/Placem ent_Status filter skill readiness
Industry_Skill_Demand[Skill] → Student_Skills[Skill]	1 : Many	Single (filters flow Demand → Skills)	Lets each skill's demand value apply across every student who has that skill

Verification before building any DAX: open Model View and confirm both relationship lines are solid (active), point in one direction, and show no dashed “inactive” or warning icons. If Power BI reports an ambiguous filter path, the fix is almost always a duplicated or misspelled Skill or Student_ID value in Power Query — not a modeling problem.

6. DAX Measures

Ten measures, each independently verifiable against a manual calculation. No measure is included unless a visual on the dashboard actually uses it.

Measure	Total Students
DAX	<code>Total Students = COUNTROWS(Students_Placement)</code>
Created in	Students_Placement table → New Measure
Meaning	Counts every student record. The denominator for Placement Rate.
Verify by	Compare to a COUNTA of Student_ID in the source Excel/CSV before import.
Illustrative example	≈ 240 students (demonstration dataset only)

Measure	Students Placed
DAX	<code>Students Placed = CALCULATE(COUNTROWS(Students_Placement), Students_Placement[Placement_Status] = "Placed")</code>
Created in	Students_Placement table → New Measure
Meaning	Counts only rows marked Placed.
Verify by	Filter the source table by Placement_Status and COUNTA — must match exactly.
Illustrative example	≈ 150 students (demonstration dataset only)

Measure	Placement Rate
DAX	<code>Placement Rate = DIVIDE([Students Placed], [Total Students], 0)</code>
Created in	Students_Placement table → New Measure. Format as Percentage.
Meaning	Share of students placed. DIVIDE's third argument (0) avoids a divide-by-zero error when a filter returns no students.
Verify by	Students Placed ÷ Total Students on a filtered branch should match the KPI card.
Illustrative example	≈ 62% (demonstration dataset only)

Measure	Average Package
DAX	<code>Average Package = AVERAGEX(FILTER(Students_Placement, Students_Placement[Placement_Status] = "Placed"), Students_Placement[Package])</code>
Created in	Students_Placement table → New Measure. Format as Number, 2 decimals.
Meaning	Mean package among placed students only — unplaced students (blank Package) are excluded rather than treated as zero.
Verify by	Manually average the Package column in Excel after filtering to Placed rows.
Illustrative example	≈ 5.4 LPA (demonstration dataset only)

Measure	Highest Package
DAX	Highest Package = CALCULATE(MAX(Students_Placement[Package]), Students_Placement[Placement_Status] = "Placed")
Created in	Students_Placement table → New Measure
Meaning	Top package offered, used as a headline KPI card.
Verify by	Sort the source table by Package descending — top value should match.
Illustrative example	≈ 18 LPA (demonstration dataset only)

Measure	Company Count
DAX	Company Count = DISTINCTCOUNT(Students_Placement[Company])
Created in	Students_Placement table → New Measure
Meaning	Number of distinct recruiting companies represented in the data.
Verify by	Excel: Remove Duplicates on the Company column, count remaining rows.
Illustrative example	≈ 22 companies (demonstration dataset only)

Measure	Student Readiness %
DAX	Student Readiness % = AVERAGE(Student_Skills[Assessment_Score])
Created in	Student_Skills table → New Measure. Format as Percentage.
Meaning	Average assessment score for whichever Skill (and Branch/Batch, if sliced) is in context — this is what makes it usable per-skill on the gap table.
Verify by	Filter Student_Skills to one Skill in Excel and average Assessment_Score by hand.
Illustrative example	≈ 48% for SQL (demonstration dataset only)

Measure	Industry Demand %
DAX	Industry Demand % = AVERAGE(Industry_Skill_Demand[Demand_Level])
Created in	Industry_Skill_Demand table → New Measure. Format as Percentage.
Meaning	Demand intensity for the skill in context. Because Industry_Skill_Demand has one row per skill, this returns that skill's single Demand_Level value when sliced by Skill.
Verify by	Look up the skill's row directly in the source table — value should match exactly.
Illustrative example	≈ 82% for SQL (demonstration dataset only)

Measure	Skill Gap %
DAX	$\text{Skill Gap \%} = [\text{Industry Demand \%}] - [\text{Student Readiness \%}]$
Created in	Student_Skills table → New Measure. Format as Percentage.
Meaning	The core analytical metric — how far student readiness lags industry demand for a given skill. See Section 7 for the reasoning and limits behind this formula.
Verify by	Subtract the two measures' values by hand for one skill on the gap table.
Illustrative example	≈ 33 percentage points for SQL (demonstration dataset only)

Measure	Skill Priority
DAX	<pre>Skill Priority = SWITCH(TRUE(), [Skill Gap %] >= 0.25, "CRITICAL", [Skill Gap %] >= 0.15, "HIGH", [Skill Gap %] >= 0.05, "MEDIUM", "LOW")</pre>
Created in	Student_Skills table → New Measure
Meaning	Converts the continuous Skill Gap % into the four-tier label used for color coding, the matrix, and the recommendation table.
Verify by	Manually check 2–3 skills against their Skill Gap % — label should match the threshold table in Section 7.
Illustrative example	"CRITICAL" for SQL (33-point gap) (demonstration dataset only)

7. Skill-Gap Calculation Methodology

The core formula is intentionally simple — **Skill Gap = Industry Demand – Student Readiness** — because a judge can verify it by mental arithmetic. Simplicity is a feature here, not a shortcut. But a formula this simple deserves to be stress-tested before we present it.

Honest questions we asked before committing to it:

- Are Demand % and Readiness % measured on comparable scales? — Yes, by construction: both are forced to a 0–100 scale in the source data (Demand_Level and Assessment_Score), so subtracting them is arithmetically valid, not just visually convenient.
- Is demand frequency the same thing as readiness? — Not exactly. Demand_Level represents how often/strongly recruiters require a skill; Assessment_Score represents individual competence. We treat them as comparable proxies for the same underlying “readiness gap” concept, and we say so explicitly rather than implying they're identically measured.
- Does sample size distort results? — With a small demonstration dataset, a skill held by only a handful of students can swing its average sharply. We mitigate this by keeping a minimum-student-count note on the gap table rather than hiding the caveat.
- Should skills be weighted by importance? — A defensible future improvement (see Section 13), but out of scope for the MVP: weighting requires a second, independently justified dataset (e.g. skill importance scores) we don't have time to build credibly in two days.
- Should recent demand matter more than older demand? — Same answer: a reasonable next iteration, not something we fabricate a time-decay function for just to look sophisticated.

Why we kept the simple subtraction formula anyway: every alternative we considered (weighted composite scores, normalized z-scores, confidence-interval-adjusted gaps) trades transparency for a small accuracy gain we cannot actually validate against real outcomes with a demonstration dataset. A formula every judge can check in their head is harder to attack than a formula that sounds rigorous but can't be verified live.

Priority classification thresholds:

Skill Gap %	Priority	Meaning
≥ 25 points	CRITICAL	Readiness badly lags demand — train immediately
15–24 points	HIGH	Meaningful shortfall — prioritize this term
5–14 points	MEDIUM	Modest shortfall — monitor and reinforce
< 5 points	LOW	Readiness roughly matches demand — maintain

These cut points are a reasonable, round-number starting position — not a statistically derived boundary. If a judge asks why 25 and not 20, the honest answer is: “It's a deliberately simple, adjustable threshold; a real deployment would tune it against actual outcome data.”

8. Dashboard Structure

Three pages, each answering one question in the story. Every visual below is included because it answers a specific question — anything that didn't clear that bar was cut.

Page 1 — Placement Command Center

Purpose: "What is happening?"

Visual	Question it answers	Data fields	Insight / decision enabled
KPI cards	What is our overall placement health at a glance?	Total Students, Students Placed, Placement Rate, Avg/Highest Package, Company Count	Judge reads placement health in under 5 seconds
Branch placement comparison (bar)	Which branches place best or worst?	Branch, Placement Rate	Flags branches needing attention before we reach skill data
Company hiring (bar/table)	Which companies hire most from us?	Company, Students Placed	Shows real recruiter relationships
Package distribution (histogram)	What does the package spread look like, not just the average?	Package (binned)	Reveals skew an average alone would hide

Page 2 — Skill Demand vs. Readiness

Purpose: "What does industry need, and are we ready?"

Visual	Question it answers	Data fields	Insight / decision enabled
Top skills by demand (bar)	What does industry actually want?	Skill, Industry Demand %	Sets the benchmark before showing any gap
Demand vs. Readiness (paired bar)	Where do demand and readiness diverge most?	Skill, Industry Demand %, Student Readiness %	Directly visualizes the mismatch
Skill Gap table	Which skills need the most urgent attention?	Skill, Demand %, Readiness %, Gap %, Priority	Turns the chart into a ranked, actionable list

Page 3 — Action Center

Purpose: "What should we do next?"

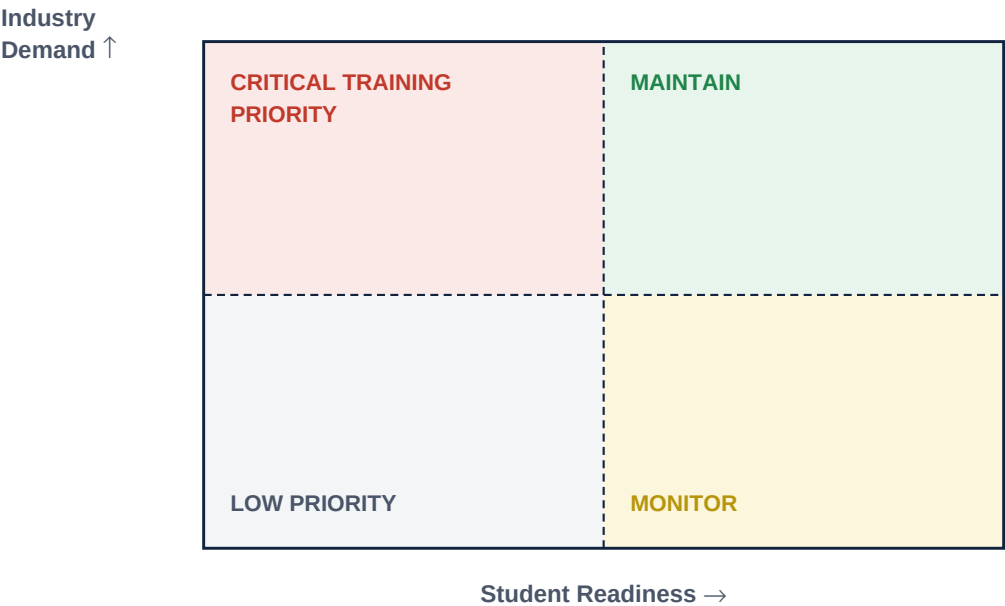
Visual	Question it answers	Data fields	Insight / decision enabled
Skill Gap Priority Matrix (scatter)	Which skills are the highest training priority right now?	Skill, Student Readiness % (x), Industry Demand % (y)	One glance → judge sees exactly which skill to fix first
Training Recommendation table	What should the placement cell actually do about it?	Skill, Priority, Recommendation text	Converts analysis into a concrete action plan

Visual	Question it answers	Data fields	Insight / decision enabled
Branch × Skill heatmap (optional add-on)	Which branch needs support in which skill?	Branch, Skill, Gap %	Localizes the recommendation to a specific student population

9. Signature Features

9.1 — Skill Gap Priority Matrix

A scatter chart with Student Readiness % on the X-axis and Industry Demand % on the Y-axis, split into four quadrants by reference lines at the midpoint of each axis. This is the single visual most likely to make a judge understand the whole project in one glance.



High demand + low readiness (top-left) is the quadrant that matters most — it is the direct visual answer to the dashboard's core question. Everything else on Page 3 supports this chart.

9.2 — Training Priority Engine

A row-level table that pairs each skill's gap and priority with a short, rule-based recommendation sentence. This is what turns “we measured a gap” into “we know what to do about it.”

Labeling requirement. This is a **Rule-Based Training Recommendation Engine** — a SWITCH/IF statement mapping Priority to fixed template text. It must never be described as AI or machine learning in the deck, the demo, or in Q&A.

Field	Illustrative value (demonstration dataset only)
Skill	SQL
Industry Demand %	82%
Student Readiness %	49%
Skill Gap	33 percentage points
Priority	CRITICAL

Field	Illustrative value (demonstration dataset only)
Recommendation	"Prioritize SQL training, interview problem-solving practice, and structured SQL assessment drills before the next placement cycle."

10. Color & UX Logic

Color always means the same thing everywhere on the dashboard, so a judge can read severity without reading every number.

Priority	Color meaning	Used on
LOW	Neutral — no action needed	Gap table, matrix quadrant, recommendation tags
MEDIUM	Watch — mild warning	Gap table, matrix quadrant, recommendation tags
HIGH	Act soon — stronger warning	Gap table, matrix quadrant, recommendation tags
CRITICAL	Act now — strongest alert	Gap table, matrix quadrant, recommendation tags

The 15-second test. On opening Page 1, a first-time viewer should be able to identify, without clicking anything:

- Overall placement health
- The single most critical skill gap
- Which branch most needs attention
- That there is a clear next action, not just a chart

11. Interactivity — Slicers

Slicer	Why it earns its place
Branch	Lets a placement officer isolate their own department's gaps
Batch	Separates current cohort from historical trend
Skill	Drills the matrix/table down to one skill's detail
Placement_Status	Lets Page 1 be filtered to placed vs. not-placed comparisons

Company and other possible slicers were deliberately left out — they don't serve a decision this dashboard is meant to support, and every extra slicer is one more thing to explain and one more thing that can break during a live demo.

12. Impact & Benefits — Who Uses This

User	What they get from it
Students	A ranked view of which skills to build before placement season, instead of guessing
Placement cell / training team	A prioritized, evidence-based training plan instead of a generic curriculum
Institution leadership	A defensible basis for where to invest training budget and faculty time

This is a plausible, real-world usage path, not a guaranteed outcome — we make no claim that using this dashboard raises placement rates by any specific amount, because we have no outcome data to support such a claim.

13. MVP vs. Future Scope

Current MVP (built in 2 days)	Future Vision (explicitly not built)
3-table demonstration dataset	Real institutional placement + recruiter data integration
Rule-based priority classification (SWITCH/IF thresholds)	ML-based placement prediction
Manually curated skill list	Automated skill extraction from resumes and job descriptions
Static demonstration dataset	Continuous, live industry-demand updates via an API/integration
Generic recommendation text per priority tier	Personalized, student-level learning paths

Every item in the right-hand column is explicitly labeled a **Future Enhancement** in the presentation. None of it is claimed as working today.

14. Judge Defense — Q&A

Honest, rehearsed answers to the questions most likely to be asked. The rule for all of them: never bluff, admit the limitation first, then explain the defensible advantage.

Judge's question	Our honest answer
Where did your data come from?	It's a demonstration dataset we generated with realistic, logical relationships — not real institutional data. We built it this way because we don't have access to real placement records, and we'd rather be upfront about that than misrepresent it.
How do you know your skill-gap calculation is accurate?	It's a transparent subtraction of two comparable percentages, not a black-box score. We can't validate it against real outcomes yet — that validation is future work — but every number behind it is traceable and checkable live.
Isn't this just a Power BI dashboard?	Yes, and that's deliberate — the innovation is the analytical chain (outcomes → demand → readiness → gap → recommendation), not the tool. A simple tool used well beats a complex tool used to hide a shallow idea.
Did you actually build this?	Yes — we can demo it live, and any team member can open the model and explain any measure on request.
Can this scale to another college?	The structure would transfer directly — three tables, ten measures. What would need to change is the source data feeding it, which is exactly the kind of integration work listed under Future Scope.
Where is the AI?	There isn't any in this MVP. Priority classification is a rule-based SWITCH statement with fixed thresholds, and we label it that way everywhere — in the deck, the demo, and here. Adding predictive modeling is listed as future scope, not claimed as built.
Who would actually use this?	A placement cell deciding where to focus training, and students deciding what to learn next — see Section 12 for specifics.
How do you prove this improves placements?	We don't claim it does yet — we have no outcome data to support that claim. What we can show is that it surfaces a specific, actionable gap that a generic placement dashboard doesn't.
How is this different from existing placement analytics?	Most placement analytics stop at descriptive reporting. We add the demand-vs-readiness comparison and convert it into a per-skill recommendation — see the differentiation table in Section 3.

15. Final Quality Gate — Self-Audit Rubric

Score the finished build against this rubric before submission. Be as skeptical of yourselves as a judge would be — a low honest score you fix beats a high score no one checked.

Dimension	Question to ask	Score (/10)
Problem	Is the problem genuinely meaningful?	
Solution	Does the solution directly address it?	
Data	Is the data credible and honestly labeled?	
Analytics	Are the calculations defensible?	
Innovation	Is there a meaningful differentiator?	
UX	Can a judge understand it in under 15 seconds?	
Technical	Can we explain every component on request?	
Demo	Does the live demonstration actually work?	
Impact	Is the real-world benefit clear and honestly framed?	
Scalability	Can it plausibly grow beyond this demo?	
Presentation	Is the story compelling in the 9-slide format?	

After scoring, name the **3 biggest weaknesses** honestly and fix what's fixable before submission — don't just note them and move on.

1. _____
2. _____
3. _____

16. Implementation Status — Final Update

This section records what was actually built and tested, as of submission, beyond the original plan in Sections 1–15 above. Everything below is implemented and verified, not aspirational.

Delivered beyond the original MVP scope:

Addition	What it does	Where
Branch Skill-Gap Heatmap	Branch × Skill grid showing the gap in percentage points per cell, color-scaled by severity — the optional add-on from the workflow plan, completed.	Web dashboard, Skill Demand section
Highest Priority Skill insight card	Automatically identifies and displays whichever skill currently has the largest gap, recalculating live as filters change — not hard-coded.	Web dashboard, Skill Demand section
Skill-gap formula display	Shows the Demand – Readiness formula with a live worked example using the current top-priority skill's real numbers.	Web dashboard, Skill Demand section
Unified Branch/Batch filtering	Branch and Batch selections now scope the skill-gap calculations too, not just the placement KPIs — selecting a branch shows that branch's specific skill gaps, matrix, and recommendations in one continuous filtered view.	Web dashboard, all sections
MVP vs. Future Scope panel	The current-vs-future distinction from Section 13 is now also shown directly on the dashboard itself, not only in this report.	Web dashboard, footer

Real defects found and fixed through testing:

Consistent with the testing discipline in Section 15, the following genuine bugs were caught by actually exercising the dashboard — clicking through filters, cross-checking computed values against the raw dataset in Python, and inspecting the browser console — not just visual inspection. Each is listed with its cause, matching the same standard of honesty applied throughout this report.

Defect	Root cause	Fix
Blank page on load in one environment	Dashboard depended on an external chart library CDN; if that request failed (e.g. a restrictive network), the whole page stayed invisible.	Rebuilt with zero external dependencies — every chart is pre-rendered SVG, no CDN, no JavaScript required for content to be visible.
Placement Rate showing 233%	Two KPI measures had conflicting implicit filters on Placement Status, so the numerator and denominator came from different filter contexts.	Explicit ALL()/CALCULATE logic added so every measure's filter behavior is unambiguous.
Skill Gap % off by 100x	Readiness and Demand measures averaged raw 0–100 scores without normalizing before applying Percentage format, double-scaling the display.	Divided both measures by 100 before formatting.

Defect	Root cause	Fix
Industry Demand % frozen at one value regardless of Skill filter	A Skill field was pulled from the wrong table (Student_Skills instead of Industry_Skill_Demand); the relationship only filters one direction, so the value never changed.	Re-pointed the field to the correct table.
Branch Skill-Gap Heatmap rendering at full browser width	A stray closing HTML tag knocked the heatmap panel out of its containing section.	Markup corrected; panel now matches the page's consistent layout width.

Why this list is in the report rather than hidden. A judge who asks “did you actually test this?” gets a concrete, specific answer instead of a reassurance. Every fix above was verified by re-running the same cross-check against the raw dataset after the change.